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ORIGINAL ARTICLE

Polysaccharides from *Arnebia euchroma* Ameliorated Endotoxic Fever and Acute Lung Injury in Rats Through Inhibiting Complement System

Ying-ye Ou,[1] Yun Jiang,[2] Hong Li,[1,3] Yun-yi Zhang,[1] Yan Lu,[2] and Dao-feng Chen[2,3]

Abstract—Arnebiaeuchroma (Royle) Johnst (Ruanzicao) is a traditional Chinese herbal medicine (TCM). It is extensively used in China and other countries for treatment of inflammatory diseases. It is known that hyper-activated complement system involves in the fever and acute lung injury (ALI) in rats. In our preliminary studies, anti-complementary activity of crude Arnebiaeuchroma polysaccharides (CAEP) had been demonstrated *in vitro*. This study aimed to investigate the role and mechanism of crude Arnebiaeuchroma polysaccharides (CAEP) using two animal models, which relate with inappropriate activation of complement system. In lipopolysaccharide (LPS)-induced fever model, the body temperature and leukocytes of peripheral blood in rats were significantly increased, while the complement levels of serum were remarkably decreased. CAEP administration alleviated the LPS-induced fever, reduced the number of leukocytes, and improved the levels of complement. Histological assay showed that there were severe damages and complement depositions in lung of the ALI rats. Further detection displayed that the oxidant stress was enhanced, and total hemolytic activity and C3/C4 levels in serum were decreased significantly in the ALI model group. Remarkably, CAEP not only attenuated the morphological injury, edema, and permeability in the lung but also significantly weakened the oxidant stress in bronchoalveolar lavage fluid (BALF) in the ALI rats. The levels of complement and complement depositions were improved by the CAEP treatment. In conclusion, the CAEP treatment ameliorated febrile response induced by LPS and acute lung injury induced by LPS plus ischemia-reperfusion. CAEP exerted beneficial effects on inflammatory disease potentially via inhibiting the inappropriate activation of complement system.

KEY WORDS: acute lung injury; fever; polysaccharides; *Arnebia euchroma*; complement.

INTRODUCTION

The complement system is an important component of the innate immunity, which takes part in the acute inflammatory response [1]. Lines of experimental evidences have shown that activation of the complement system plays a major role in the febrile response [2, 3] and acute lung injury [4, 5]. Complement inhibitors with appropriate complement modulatory activity can lead to a novel treatment strategy for various diseases [6, 7].

Fever, most manifest symbol of infectious illness, is one of concatenations of complex and nonspecific host defense responses to infections [8]. It is known that hyper-activated complement system involves in fever [9].

Ying-ye Ou and Yun Jiang contributed equally to this work.

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[1] Department of Pharmacology, School of Pharmacy, Fudan University, Shanghai, 201203, China
[2] Department of Pharmacognosy, School of Pharmacy, Fudan University, Shanghai, 201203, China
[3] To whom correspondence should be addressed to Hong Li at Department of Pharmacology, School of Pharmacy, Fudan University, Shanghai, 201203, China. E-mail: lxzhang@shmu.edu.cn; and Dao-feng Chen at Department of Pharmacognosy, School of Pharmacy, Fudan University, Shanghai, 201203, China. E-mail: dfchen@shmu.edu.cn

Acute lung injury (ALI)/acute respiratory distress syndrome (ARDS) is uncontrollable systemic inflammatory responses following trauma [10]. The extensive host tissue damage is sustained and amplified by unrestricted complement activation [7]. The inappropriate activation of complement system might relate with capillary leakage, which can result in hypoxemia, pulmonary edema, and oxidative stress damage in ALI [11].

As a prototypical endotoxin, lipopolysaccharide (LPS) has been widely used to study the acute inflammatory response, such as febrile response [12], sepsis, and acute lung injury (ALI) [13, 14]. The presence of LPS in blood causes the immediate activation of the complement cascade. The generation of anaphylatoxins C5a boosts the inflammation procedure, which simulates the PGE2 production [6, 9]. PGE2 stimulates vagal afferents and provides the signal mediating the febrile response to the brain [6, 9]. In animal models of acute lung injury, ischemia-reperfusion plus LPS can induce damage via systemic activation of the complement system and related events [15, 16].

The root of *Arnebiaeuchroma* (Royle) Johnst (Ruanzicao) is a traditional Chinese herbal medicine. It is extensively used in China and other countries for treatment of inflammatory diseases [17] and cancer [18]. The dried root of *A. euchroma* has been reported to be used externally to treat toothache, earache, and skin eruptions [19]. Previous screening studies have reported that shikonin and alkannin from Ruanzicao and other species demonstrated anti-inflammatory activity [20, 21], as well as their derivatives. However, the complement modulatory properties of crude *Arnebiaeuchroma* polysaccharides (CAEP) have not been elucidated.

In our preliminary studies, anti-complementary activity of CAEP had been demonstrated *in vitro*. According to the result from the preliminary experiments, we hypothesized that CAEP ameliorated febrile response and acute lung injury in rats via its modulatory effect on complement system.

MATERIALS AND METHODS

Animals

Male Sprague Dawley (SD) rats (250 to 300 g) were purchased from Super-B&K Animal Ltd. (SPF II, Certificate No. SCXK2008-0082), and male Wistar rats (210 ± 20 g) were purchased from Slaccas-Shanghai Lab Animal Ltd. (SPF II, Certificate No. SCXK2007-0005).

Reagents

LPS (*Escherichia coli* 0111:B4 endotoxin) was from Sigma-Aldrich. Myeloperoxidase (MPO), malondialdehyde (MDA), and superoxide dismutase (SOD) kits were purchased from Nanjing Jiancheng Biotechnology Co., Ltd. (Nanjing, Jiangsu, China). Complement components C3 and C4 kits were purchased from Taiyang Biotechnology Co., Ltd. (Taiyang, Shanghai, China). All other reagents were of the highest quality available.

Plant material and extract preparation and characterization of crude *A. euchroma* polysaccharides (CAEP) are available in the online-only Supplement Material.

Complementary Activity Assay (Classical Pathway)

The complement activity was measured according to the Mayer's modified method [22]. Sensitized erythrocytes (EAs) were prepared by incubation of 2% sheep erythrocytes with rabbit anti-sheep erythrocyte antibody (1:400) in isotonic veronal-buffered saline (VBS^{2+}) (containing 0.5 mM Mg^{2+} and 0.15 mM Ca^{2+}) in equal volumes. Samples were dissolved in VBS^{2+}. Guinea pig serum was used as the complement source. The 1:10 diluted Guinea pig serum was chosen to give submaximal lysis in the absence of complement inhibitors. In brief, various dilutions of the sample (200 μL) were mixed with Guinea pig serum and EAs for the equal volume, and then the mixture was incubated at 37°C for 30 min. Assay controls were incubated in the same conditions, including (1) 100% lysis: 200 μL EAs in 400 μL ultrapure water; (2) 0% lysis: 200 μL EAs in 400 μL VBS^{2+}; and (3) samples background: 200-μL dilution of each sample in 400 μL VBS^{2+}. The mixtures were centrifuged immediately at 4°C after incubation. Optical density of the supernatants was measured at 405 nm on well scan (Thermo Scientific Multiskan FC). The percentage of hemolytic inhibition was calculated as follows: inhibition of lysis (%) $= 100 - 100 \times (OD_{sample} - OD_{sample\ background} - OD_{0\%\ lysis}) / (OD_{100\%\ lysis} - OD_{0\%\ lysis})$.

Fever

Model and Grouping

Adult male Wistar rats weighing 200 ~ 230 g were used. The animals were housed at 22 ± 2°C under a 12-h light–dark cycle at least 3 days before experiment. The rats were housed in two of one cage, after fasting 10 h with free access to water. The basal body temperature was determined three times by thermometer before experiment, and then the rats with unstable temperature were rejected. There were six groups ($n = 6$) in this experiment: sham group, LPS-induced model group, CAEP (at different doses of 50, 100, and 150 mg/kg) groups, and aspirin (150 mg/kg)-treated group. The drugs were diluted in normal saline for oral administration. 30 min after drug administration; LPS (60 μg/kg) was injected intraperitoneally to induce fever. The same volume saline (1 mL/kg) without LPS was injected in the sham group. The body temperature was measured once every 30 min from 0 to 6 h after LPS injection. The change of body temperature was calculated ($\triangle T = \text{body temp.}_{\text{at the time}} - \text{body temp.}_{\text{basic}}$). The responses of each rat were evaluated by areas under temperature–time curve, accordingly obtaining temperature response indexes (TRI, °C × h). After the rats were anesthetized with intraperitoneal injection of 1 g/kg urethane, blood sample was collected for leukocyte counts, and serum was collected and stored at −80°C for the complement assay.

Hemolysis of Serum and the Total Number of Leukocytes in Peripheral Blood

The complement activity was measured according to the protocol described above. The collected serum was diluted at 1:10 VBS^{2+} buffer solution for hemolytic assay. Hemolysis (%) = $100\% \times (OD_{sample} - OD_{0\%\ lysis}) / (OD_{100\%\ lysis} - OD_{0\%\ lysis})$. The total number of leukocytes in peripheral blood was detected on the Neubauer's counting chamber.

Acute Lung Injury

Model and Grouping

The CAEP were dissolved in normal saline for oral administration. The rats were subjected to both ischemia-reperfusion (IR) injury and LPS injection to induce "two-hit" ALI model. The rats were randomly divided into six groups ($n = 8$ each group): sham group, ALI model group, CAEP (at doses of 40, 80, and 120 mg/kg, ig) group, and hydroprednisone (70 mg/kg, ip)-treated group. Induction of ALI in the rats was performed as described in our previous studies with some modification [23]. Male SD rats were anesthetized with intraperitoneal injection of 1 g/kg urethane. The right carotid artery was cannulated for monitoring the mean arterial pressure (MAP). Hemorrhagic shock lasted for 60 min with the MAP was maintained to 40 ± 5 mmHg. Blood sample was added with 4% heparin (0.1 mL) to prevent clotting. After a hypotensive period of 60 min, the animals were resuscitated by transfusion of the shed blood over a period of 1 h. Drug or saline was administered 30 min after the beginning of resuscitation. The sham animals underwent the same surgical procedures, but hemorrhage was not taken. At the end of resuscitation, a tracheotomy was operated, meanwhile either LPS (1.5 mg/kg in about 200 μL saline) or saline alone (sham group) was dripped intratracheally.

Serum was collected at 2 and 4 h after LPS instillation and stored at −20°C for complement measurement. The experiment was ended 4 h after LPS instillation. Other samples were collected as described above.

Lung Wet/Dry Weight Ratio

The lower lobe of the right lung was excised and weighted as wet weight. The lung was then placed in an incubator at 80°C for 48 h to obtain the dry weight. The wet/dry weight ratio (W/D weight ratio) was calculated by the wet weight dividing by the dry weight.

Bronchoalveolar Lavage Fluid Collection and Cell Counts

Following removal of the lung's lower right lobe, bronchoalveolar lavage fluid (BALF) was collected by flushing 2 mL normal saline into the left lung via a tracheal cannula (four times). The pooled BALF was centrifuged at 111×*g* at 4°C for 10 min. Pelleted cells were then resuspended in 1 mL of saline. Total cell number was counted by hemacytometer. The supernatant of the lavage was collected and stored at −20°C for further measurement.

Protein Concentration, MPO Activity, MDA Concentration, and SOD Activity in BALF

Total protein concentration in the supernatant of BALF was determined with well scan (Thermo Scientific Multiskan FC) by measuring the absorbance at 540 nm after the addition of Coomassie brilliant blue [24]. As an index of neutrophil infiltration, MPO activity in BALF was determined by MPO kits. The concentrations of MDA and the SOD activity in BALF were measured using commercially available colorimetric assay kits.

Histological Examination and Immunohistochemistry

Superior lobes of the right lung were immediately fixed in 10% formalin, embedded in paraffin, sectioned in 5-μm thickness, and stained with hematoxylin and eosin

(H&E). Complement deposit was determined using immunohistochemical staining with rabbit anti-human C3c (two-step anti-rabbit detection reagent (HRP) kit, Antibody Diagnostic Inc) [5].

Serum Total Component Hemolytic Activity and C3 and C4 Levels

Serum was collected at 2 or 4 h after LPS instillation and diluted at 1:30 VBS^{2+} for hemolytic assay. The contents of complement components C3 and C4 were measured using commercially available kits.

Statistical Analysis

Data are presented as mean ± SD. Differences between the groups were analyzed using one-way ANOVA and Dunnett's test. $P < 0.05$ was considered significant.

RESULTS

Chemical Characterization of CAEP

CAEP showed three major peaks in the chromatogram of high-performance gel permeation chromatography (HPGPC) with several minor ones (Online Resource Fig. S_1). GC analysis indicated that CAEP was mainly composed of Ara, Glc, GaL, Man, and Rha in the ratio of 3.69:2.57:1.47:1.20:1, together with trace amounts of Fuc and Xyl (Online Resource Fig. S_2). The total carbohydrate and uronic acid contents of CAEP were 76.36 ± 0.93 and 24.37 ± 1.56%, respectively ($n = 3$). The total protein content was 9.38 ± 1.85% ($n = 3$).

In Vitro Anti-complementary Effect

CAEP significantly inhibited hemolysis in sheep erythrocytes induced by Guinea pig serum ($CH_{50} = 0.397 \pm 0.011$ mg/mL) (as shown in Fig. 1).

Lysis of sheep erythrocytes in 1:30 diluted Guinea pig serum in the presence of increasing amounts of CAEP was observed. Data are listed as mean ± SD ($n = 3$).

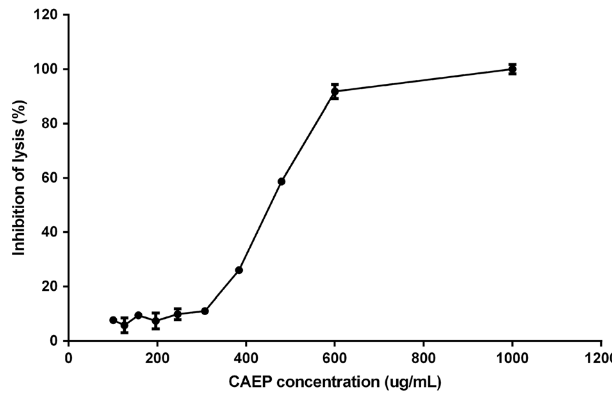

Fig. 1. Inhibition of the classical pathway-mediated hemolysis by CAEP.

Effect of CAEP on LPS-Induced Febrile Rats

Effect of CAEP on Body Temperature and the Temperature Response Index in the LPS-Induced Febrile Rats

During the experiment, the sham group rats' body temperature was maintained at a normal range with minor changes. The body temperature and temperature response index (TRI) of the model group were significantly higher than those of the sham group after the LPS stimulation ($P < 0.001$) (Fig. 2a).

As shown in Fig. 2a, the CAEP (100 and 150 mg/kg) treatment significantly inhibited the fever induced by LPS from 30 min to 6 h ($P < 0.001$). Aspirin also inhibited the temperature increase. Similar results were found in temperature response index (Fig. 2b).

Effect of CAEP on Complement Hemolytic Activity of Serum and Number of Leukocytes in Peripheral Blood in the LPS-Induced Febrile Rats

As shown in Fig. 2c, complement levels of the model group were significantly decreased compared with those of the sham group ($P < 0.001$). LPS injection increased the number of leukocytes in peripheral blood ($P < 0.05$) in the model rats (Fig. 2d).

The CAEP (100 and 150 mg/kg) treatment partly restored complement levels in serum ($P < 0.01$) and the number of leukocytes in peripheral blood (CAEP 150 mg/kg, $P < 0.05$). No significant difference was observed between the aspirin treated group and LPS-induced model group.

Effect of CAEP on ALI

Effect of CAEP on Lung Pathology

Histopathological examination confirmed existence of obvious lung inflammation and injury in the ALI model rats treated with vehicle, but much less severe abnormalities (lung injury and edema morphologically) were observed in the CAEP- and hydroprednisone-treated groups (Fig. 3a).

Higher levels of complement deposition were present in the model group than those in the sham group.

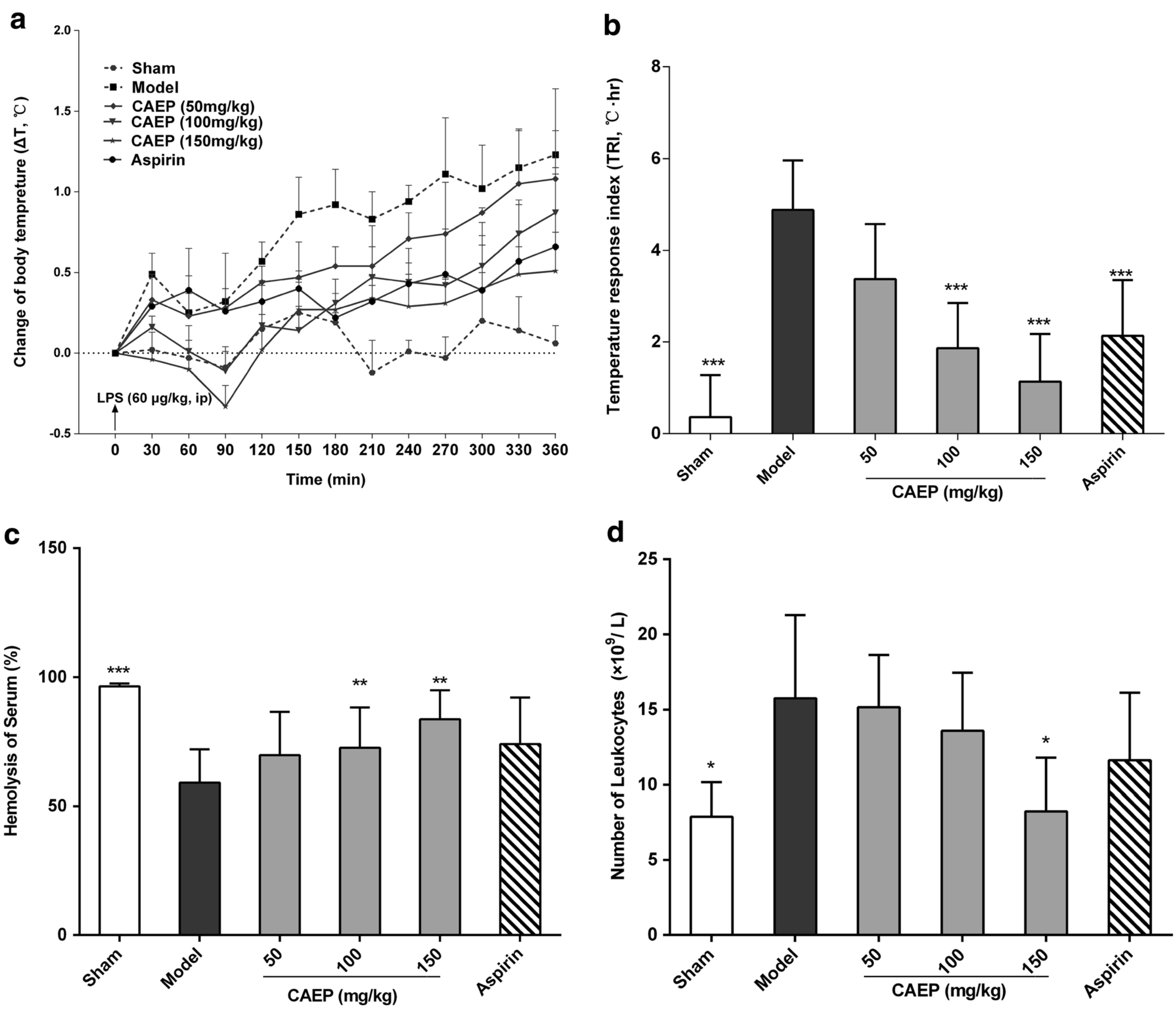

Fig. 2. Effect of CAEP on LPS-induced febrile rats. **a**, **b** The change of body temperature and the temperature response index. **c** Serum total hemolytic activity. **d** Count of leukocytes in peripheral blood. Data were expressed as mean ± SD; $n = 6$. *$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$ compared with the model group, tested by ANOVA and Dunnett's test.

Treatment with CAEP and hydroprednisone significantly reduced the complement deposition induced by ischemia-reperfusion plus LPS (Fig. 3b).

CAEP Attenuated Severity of Inflammation in the Lung and in BALF

As shown in Fig. 4a, the lung W/D ratio was significantly higher in the ALI group than that in the sham group ($P < 0.001$). This increase was significantly inhibited by the CAEP (80 and 120 mg/kg) treatment as well as hydroprednisone ($P < 0.05$).

Protein concentration and the number of leukocytes were elevated in BALF in the ALI group ($P < 0.001$) compared with those in the sham group. Treatment with CAEP (80 and 120 mg/kg) and hydroprednisone (70 mg/kg) significantly reduced the protein level and the number of leukocytes of BALF ($P < 0.001$) (Fig. 4b, c).

In Fig. 4d, results indicated that MPO activity was significantly higher in the ALI model group than that in the sham group ($P < 0.001$). Treatment with the CAEP at the dose of 120 mg/kg reduced the MPO activity in BALF compared with the model group ($P < 0.05$). Treated with 80 mg/kg CAEP or hydroprednisone only had a slight decrease in MPO activity without significance.

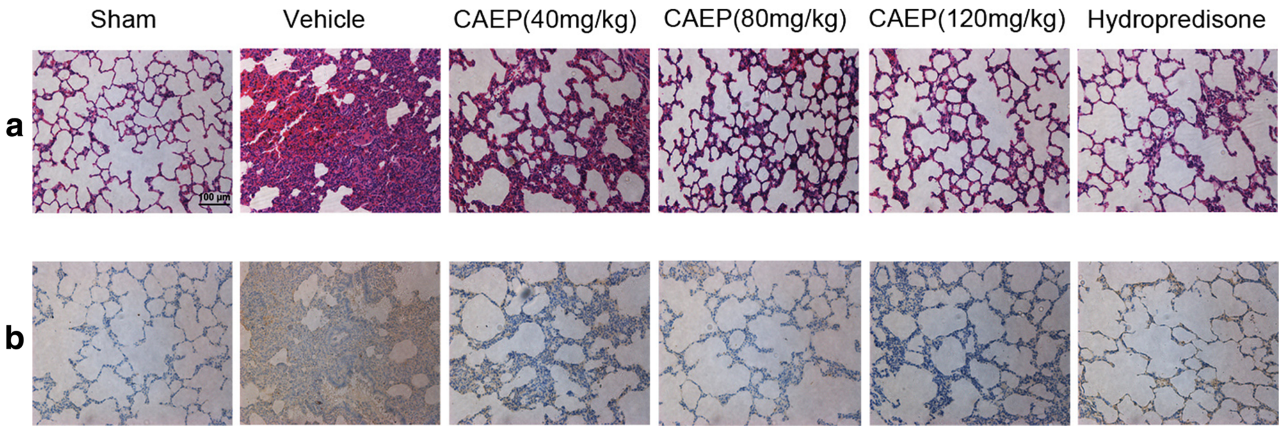

Fig. 3. Effect of CAEP on lung pathology in the ALI rats. The upper lobe of the right lung was fixed in 4% formalin and 5-μm sections were prepared. **a** Stained with hematoxylin and eosin. **b** immunochemistry microscopic images of the lung of each group indicated the complement deposition. Representative graphs of the rats in each group. Original magnification, ×200.

CAEP Improved the Injury of Oxidative Stress in the ALI Rats

High levels of oxygen-free radical directly damaged lung tissues. The protective effect of CAEP on oxidative stress was evaluated by measuring the SOD activity and MDA levels.

A significantly lower SOD activity was observed in BALF of the ALI model group compared with that of the sham group ($P < 0.001$). CAEP (120 mg/kg) treatment improved the SOD activity significantly ($P < 0.01$) (Fig. 4e).

The result demonstrated that the level of MDA was reduced by the CAEP (80 and 120 mg/kg) and hydroprednisone treatments ($P < 0.05$), compared with that by the saline-treated sham and model groups (Fig. 4f).

Effect of CAEP on Total Component Hemolytic Activity and Levels of C3 and C4 in Serum

In the ALI model group, the hemolytic activity of serum was significantly decreased at 2 and 4 h after LPS injection ($P < 0.05$). The CAEP treatment improved the complement consumption. There was no significantly difference between the hydroprednisone-treated group and the ALI model group (Fig. 5a).

Four hours after LPS challenging, the serum levels of C3 and C4 were significantly decreased in the ALI group ($P < 0.001$). The CAEP (120 mg/kg) and hydroprednisone treatments inhibited the decrease of C3 levels in serum ($P < 0.05$), and all the doses of CAEP and hydroprednisone significantly attenuated decrease of C4 levels in the ALI rats ($P < 0.05$) (Fig. 5b).

Serum was collected 2 and 4 h after LPS injection and diluted at 1:30 in VBS^{2+} for hemolytic assay. Serum was collected 4 h after LPS injection and diluted at 1:2 in normal saline (NS) for the assays of complement components C3 and C4, using the immunonephelometry kits. Data were expressed as means ± SD ($n = 8$). $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$ compared with those in the ALI model group, tested by ANOVA and Dunnett's test.

DISCUSSION

The root of *Arnebiaeuchroma* (Royle) Johnst is one of the most popular plants in Chinese medicine as Ruanzicao [18]. The recent studies about *A. euchroma* focus on the pharmacological activity of shikonin [25, 26]. It has been reported that hydroxynaphthoquinone fraction from *A. euchroma* possesses promising therapeutic properties for rhinitis [17]. In this study, the anti-complement properties of polysaccharides isolated from *A. euchroma* root were demonstrated *in vitro* and *in vivo*.

Fever is a common clinical symptom in many infectious diseases [27]. The LPS-induced fever is characterized by the higher body temperature accompanied with leukocyte increasing and complement activation [28]. It had been reported that pyrogenic LPS may rapidly activate the complement cascade and that one or more of the fragments generated could contribute importantly to the initiation of the febrile response [9].

In this study, LPS challenging significantly increased the body temperature and leukocytes in peripheral blood with remarkably decreased complement reaction. CAEP

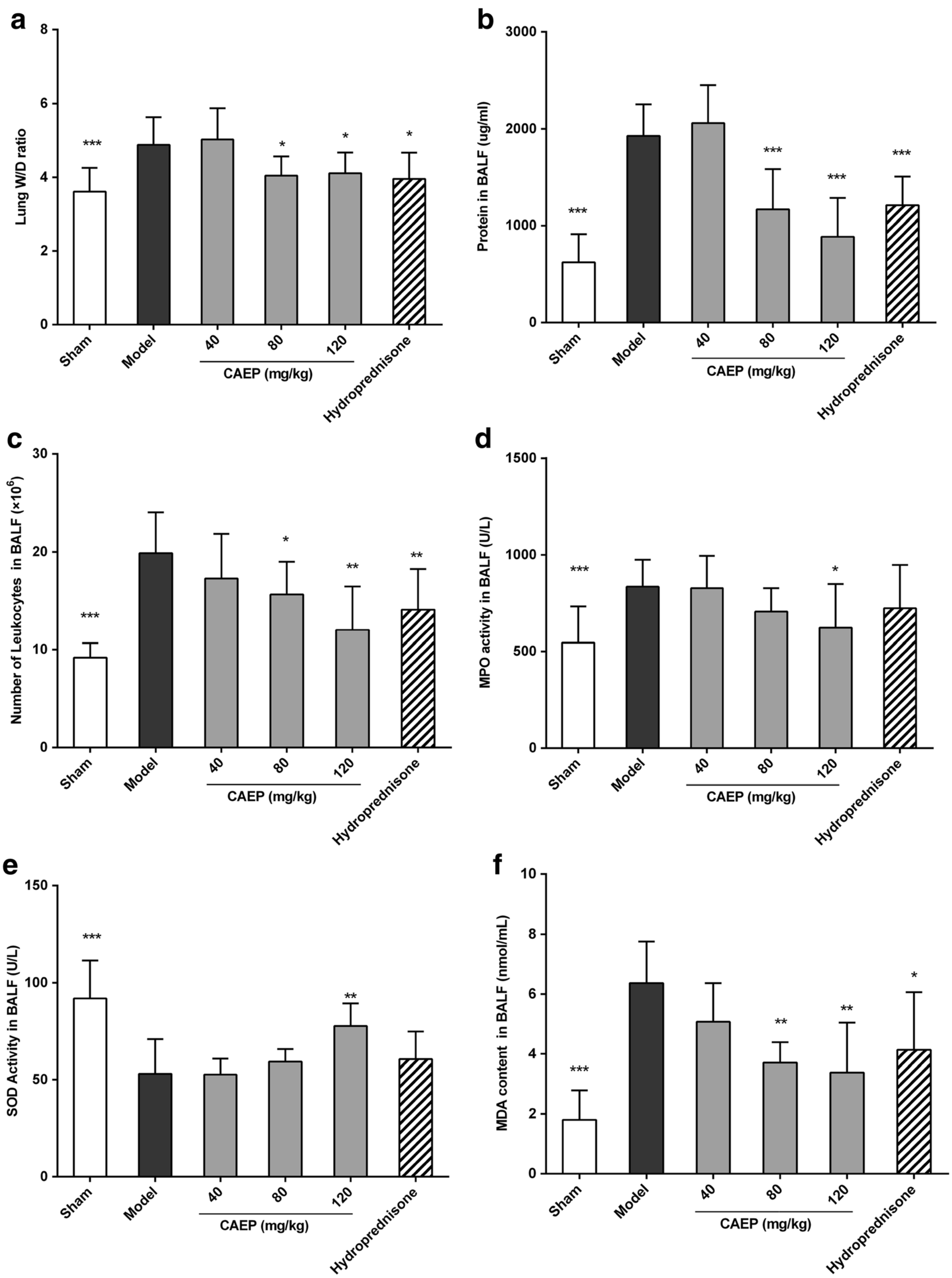

Fig. 4. CAEP protected against lung injury. **a** The lung W/D ratio. **b** Protein concentrations in BALF. **c** Number of leukocytes in BALF. **d** MPO activity in BALF. **e** SOD activity in BALF. **f** Levels of MDA in BALF. The values are expressed as the mean ± SD ($n = 8$); $*P < 0.05$, $**P < 0.01$, and $***P < 0.001$ compared with the ALI model group, tested by ANOVA and Dunnett's test.

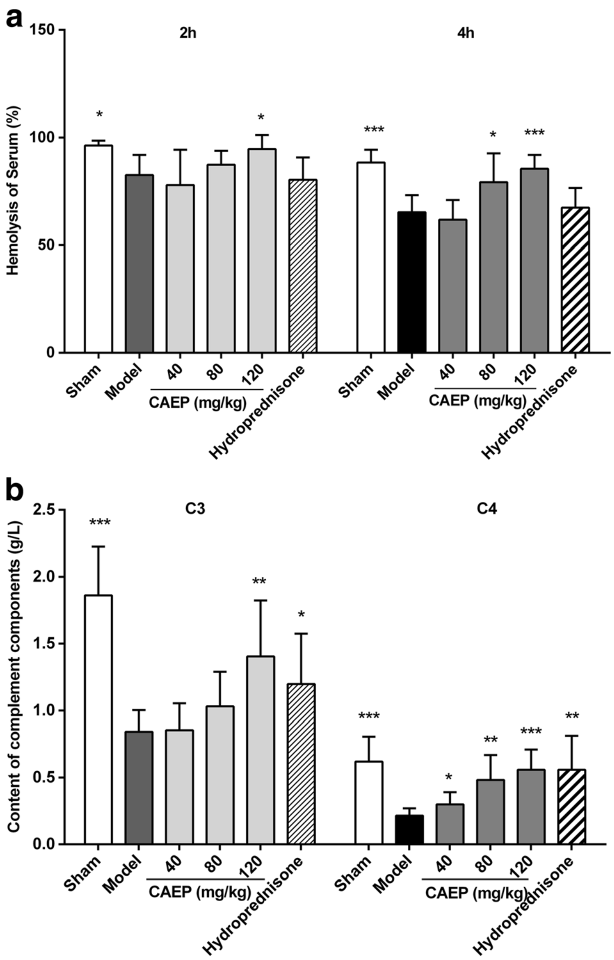

Fig. 5. Effect of CAEP on complement activity in the ALI rats.

with *in vitro* anti-complementary property alleviated all these symptoms.

In this experiment, the rats were subjected to both ischemia-reperfusion (IR) injury and LPS injection. This "two-hit" model is recommended by the National Heart, Lung, and Blood Institute of America [29]. The use of hemorrhagic shock prior to the introduction of intratracheal endotoxin produced significant increases in neutrophil infiltration as well as protein leakage across the lung epithelial barrier, when as compared to endotoxin alone [23]. It has been established that IR injury can lead to lung damage via systemic activation of the complement system and related events [16]. Lines of experimental evidence have shown that animals subjected to intestinal I/R injury, which were treated with complement inhibitors, showed attenuation of tissue injury [15].

In the lungs of the ALI model rats, HE section displayed serious pathological injury, such as lung congestion and edema, increased vascular permeability, and extensive neutrophil infiltration. The immunohistochemistry results showed abundantly complement deposition indicating over-activation of complement.

The above damages were ameliorated by the CAEP treatment. After oral administration, CAEP effectively improved the lung W/D ratio, protein leakage, leukocytes infiltration, and MPO activity. MPO, released by neutrophils, related close with the activation, adhesion, and recruitment of the cells. MPO is critically involved in the induction of organ damage after ischemia-reperfusion [30]. CAEP administration alleviated the large influx of neutrophils and inhibited the MPO activity, which reduced lung injury.

MDA, the ultimate product of unsaturated lipid peroxidation, reflected the content and extent of the radical lipid peroxidation [31]. SOD is one of the most important endogenous scavenging antioxidant enzymes. The consumption and depletion of SOD could lead to increased production of reactive oxygen species [32]. To investigate the effect of CAEP on oxidative stress, we analyzed the levels of MDA and the activities of SOD in BALF. The levels of MDA were significantly higher in the ALI model group than those in the sham group. The rats challenged with ischemia/reperfusion and LPS showed lower activities of SOD in the BALF. The CAEP treatment significantly decreased the MDA levels and increased SOD activities in BALF. Administration of CAEP partly improved the oxidative stress in the ALI rats.

The complement system is part of the innate immune system, which plays an important role in host defense and immune homeostatic [11]. The complement system is activated in three pathways: the antibody-dependent classical pathway, the antibody-independent alternative pathway, or the mannose-binding lectin/mannose-binding lectin-associated serine protease (MBL/MASP) pathway [7]. In the classical pathway and MBL/MASP pathway, the cleavage of complement factors C4 and C2 and subsequent activation of C3 started the complement cascade. The alternative pathway activated by the forming of stable C3 convertase [9, 15].

But excessive or inappropriate activation of complement also contributes to the development of many inflammatory and immunological diseases [3, 33–35]. During acute pathologic damages, such as sepsis or ischemia-reperfusion [1, 36], unrestricted complement activation easily can overcome the protection of the complement regulators and may result in extensive host tissue damage [15]. Excessive or prolonged complement activation can arise from these conditions with complement component exhaustion [7].

In this experiment, the total hemolytic activity and C3 and C4 levels were decreased in the ALI model group when compared with those in the sham group. After the CAEP treatment, the content of C3 and C4 in serum and the activity of the total hemolytic activity were improved. The CAEP treatment also reduced the complement deposition in the lung of the ALI animals, which may be helpful in alleviating morphological injury.

It has been confirmed that oxidative stress results in complement activation and deposition on the vascular endothelium [37]. Excessive activation of complement recruits neutrophils that results in tissue damage [38]. Prevention of complement activation following oxidative stress may greatly decrease harmful injury, subsequent inflammation, and tissue damage [37]. In this study, CEAP demonstrated a beneficial effect on inhibiting the activity of complement *in vitro* and *ex vivo*, and reducing oxidative level in the ALI rats.

It is reported that pharmacological activity of polysaccharide may be affected by various factors, including monosaccharide composition, glucosidal bonds, molecular size, branching degree, and the overall molecular conformation [39]. In this study, CAEP was crude polysaccharide with three major peaks in HPGPC and several minor ones. The influence of some factors could not be explicated. Homogeneous polysaccharides possess definite characterization. Thus, preparation of homogeneous polysaccharides from CAEP and study of modulatory effect on complement system will be performed in the future.

In conclusion, our study suggested that the crude polysaccharide was one of the beneficial components of *A. euchroma*. The CAEP treatment ameliorated LPS-induced febrile response and acute lung injury in the rats. The underlying mechanism was related to its modulatory effect on excessive activation of complement system.

CAEP, Crude *Arnebiaeuchroma* polysaccharides; ALI, Acute lung injury; LPS, Lipopolysaccharide; NS, Normal saline; BALF, Bronchoalveolar lavage fluid; CP, Classical pathway; EAs, Sensitized erythrocytes; VBS^{2+}, Veronal-buffered saline; MAP, Mean arterial pressure; SOD, Superoxide dismutase; MDA, Malondialdehyde; MPO, Myeloperoxidase

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Compliance with Ethical Standards. The use of animals was approved by the Animal Ethical Committee of School of Pharmacy of Fudan University (Ethical Review No. 2011-9).

Conflict of Interest. The authors declare that they have no conflict of interest.

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